

UniFaRN: Unified Transformer for Facial Reaction Generation

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ABSTRACT

We propose the **Unified Transformer for Facial Reaction Generation** (UniFaRN) framework for facial reaction prediction in dyadic interactions. Given the video and audio of one side, the task is to generate facial reactions of the other side. The challenge of the task lies in the fusion of multi-modal inputs and balancing appropriateness and diversity. We adopt the Transformer architecture to tackle the challenge by leveraging its flexibility of handling multi-modal data and ability to control the generation process. By successfully capturing the correlations between multi-modal inputs and outputs with unified layers and balancing the performance with sampling methods, we have won first place in the REACT2023 challenge.¹.

CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → *Human computer interaction (HCI)*.

KEYWORDS

facial reaction prediction, dyadic human-agent interaction

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1 INTRODUCTION

With the success of Large Language Models (LLMs), we are running into an era of intelligent agents, where the demand for naturally behaving agents is soaring. However, the problem of automatically generating an appropriate response from the agent

when interacting with humans still lacks perfect formalization and thorough investigations. Past works investigating this problem [6][18][19][20][25] mostly take only verbal inputs [19] or neglect the one-to-many nature of appropriate reaction mappings [25]. In application, ignoring non-verbal inputs and repeating behaviour when teased by users significantly impact the realism of the agent.

The REACT2023 challenge [23] first formalizes the Multiple Appropriate Reaction Generation (MARG) problem, where the algorithm generates various appropriate reactions to the same context. Specifically, given a speaker’s dyadic interaction video and audio, we generate multiple sequences of the listener’s facial reactions, comprising Facial Action Units (AUs), Valence and Arousal (VA), and facial expressions. The challenge includes offline prediction from whole 30-second inputs and online, frame-by-frame prediction. The difficulties lie in cross-modal mapping from video and audio to labels, and balancing appropriateness with diversity.

To tackle this new challenge, we propose a multi-modal framework named **Unified Transformer for Facial Reaction Generation** (UniFaRN), which features unifying multi-modal tokens and unifying online and offline tasks. Specifically, we embed video, audio, and labels into facial, pose, acoustic, and reaction tokens, which are fused uniformly by the Transformer network as a single sequence. In each Transformer block, we further unify both online and offline tasks by the shared fast-forward layer, while we navigate the tokens through causal and bi-directional self-attention layers for the two tasks respectively. The tokenization of labels enables us to balance appropriateness with diversity by sampling. The experimental results show the effectiveness of the UniFaRN, resulting in ranking first in the REACT2023 challenge.

2 METHOD

Let $\mathcal{V} = \{I_1, I_2, \dots, I_T\}$ denotes the speaker video with T frames, we divide the corresponding audio $\mathcal{A}$ into windows of 40ms length each for aligning to video frames, denoted as $\{a_1, a_2, \dots, a_T\}$. Given $\mathcal{V}$ and $\mathcal{A}$ of the speaker in a dyadic interaction, the task is to predict facial reaction sequence $\mathcal{R} = \{r_1, r_2, \dots, r_T\}$ of the listener, in which the reaction frame $r_t = \{au_t^1, \dots, au_t^{15}, valence_t, arousal_t, exp_t^1, \dots, exp_t^8\}$ consists of AUs, VA, and expressions.

The architecture of the UniFaRN is illustrated in Figure 1. Utilizing the same Transformer stem for both tasks, we map multi-modal features into tokens, add modality embeddings, and then unify them in the Transformer via modality-agnostic self-attention layers. According to BLIP [8], sharing parameters in causal and bi-directional self-attention layers degrade performance, while sharing feed-forward layers improves learning. Following such insights,

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¹<https://sites.google.com/cam.ac.uk/react2023>

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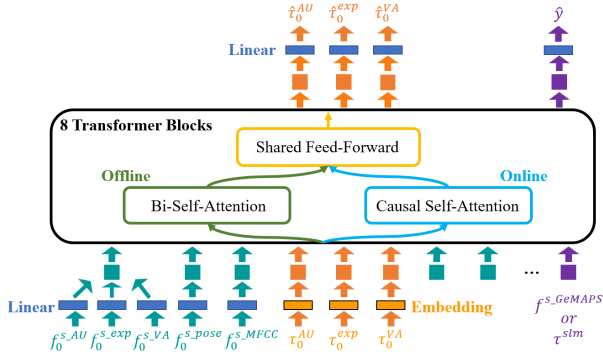


Figure 1: Architecture of the UniFaRN.

we navigate both tasks through different self-attention layers and then unify them in shared feed-forward layers. The output reaction tokens $\hat{\tau}$ are predicted by linear layers. The whole process spans three stages: pre-processing, training, and inference, the ideas and details of which are explained as follows.

2.1 Pre-processing

By intuition, raw speaker video often contains extraneous information like background and clothing, which can be more prominent in the pixel space than reaction-related features such as facial movements. These unnecessary features can overload the network.

To focus our model on learning mappings of reactions, we leverage off-the-shelf tools to extract reaction-relevant features, limiting video input to facial and pose features. We also preprocess the audio for the network. Thus, the speaker input consists of three types of features: facial, pose, and acoustic, as per Equation 1.

$$F = \{f_1^S, f_2^S, \dots, f_T^S\} \cup f^{s_GeMAPS}_T \quad (1)$$

$$f_t^S = \{f_t^{s_facial}, f_t^{s_pose}, f_t^{s_MFCC}\}$$

2.1.1 Facial Features. Since the targets of the task are AUs, VA, and expression, we consider features of these three to be the most relevant, as formulated in Equation 2.

$$f_t^{s_facial} = \{f_t^{s_AU}, f_t^{s_exp}, f_t^{s_VA}\} \quad (2)$$

Prior to extracting facial features, we crop the face region from the video using OpenCV and Dlib tools. We return the cropped face once detected by either tool, or a zero tensor with a faceless flag if no face is detected. The AUs features $f_t^{s_AU}$ and expression features $f_t^{s_exp}$ are the output of the penultimate layer of ME-GraphAU[12] and ResMaskNet[15], respectively. We extract VA features $f_t^{s_VA}$ by FaceTorch². The dimensions of the extracted AUs, VA, and expression features are 25088, 1408, and 512, respectively.

2.1.2 Pose Features. By manually validating the given dataset, we find that some samples have extreme poses or are highly obscured, making the facial features all absent. Therefore, we supplement the learning with pose features using VideoMAE [24] as our extractor, which can derive action-related features from frame sequences. Using a 5-frame window, we derived 150 pose features from each

²<https://pypi.org/project/facetorch/>

sample's 750 video frames. We then repeated the features to match the frame number. The dimension of each pose feature is 1408.

2.1.3 Acoustic Features. Following previous work [16], we embed audio into Mel Frequency Cepstral Coefficients (MFCCs) and Geneva Minimalistic Acoustic Parameter Set (GeMAPS) [4] features. The MFCCs are aligned with video frames, while the GeMAPS features are integrated from the whole 30-second audio. The GeMAPS features include LLD and Functionals features, as per Equation 3.

$$f^{s_GeMAPS}_t = \{f_t^{s_GeFun}, f_t^{s_GeLLD}\} \quad (3)$$

First, we use the FFMPEG tool to extract audio from the MP4 files at 2 channels and 192 kbps. Then, we obtain MFCCs and GeMAPS features by the python-speech-feature[13] and OpenS-MILE [5] tools, respectively. For MFCCs, we employ a 176375 Hz sampling rate, 0.04-second window size and stride, and 26 filter banks, yielding 25 frames/sec and feature shape $f_t^{s_MFCC}$ of (750, 26). For GeMAPS, we separately extract LLD and Functionals, resulting in a (2996, 65)-shape tensor and a 6373-dimension vector, respectively.

2.1.4 Tokenization. For the advantage of controllable sampling to find a balance between appropriateness and diversity, we propose to tokenize the listener labels. Before tokenization, the listeners' AUs, VA and expression are 15 {0, 1} values, two (-1, 1) values, and 8 (0, 1) probabilities, respectively. We tokenize the AUs, VA and expression into tokens τ_t^{AU} , τ_t^{exp} , and τ_t^{VA} by separate codebooks, resulting in vocabularies of 5976, 2710, and 8000, respectively.

Specifically, we tokenize AUs by identifying the most common combinations of all 15 AUs' activation per frame, assigning these combinations unique identifiers in the AU codebook. VA values are tokenized by dividing the Valence and Arousal 2D space into 80×80 grids, mapping VA values to the nearest grid point, which are then assigned identifiers in the VA codebook. Expression probabilities are tokenized using K-means to group the 8-dimensional vectors into 8000 classes. Each class is assigned an identifier to index the expression codebook, with the class centers as the elements.

2.2 Training

We randomly sample one reaction sequence from all appropriate reactions to the speaker as the label in each iteration.

2.2.1 Offline Task. During offline training, we utilize bi-directional self-attention to condition the prediction on entire speaker inputs, leaving causal self-attention layers inactive. Here we enforce the **Masked Reaction Modeling (MRM) Loss**, L_{mrm} , as the main constraint. We append the GeMAPS features $f^{s_GeMAPS}_t$ to the input sequence. Similar to Masked Language Modeling [7], we randomly mask out listener tokens while leaving speaker tokens intact. The model has to predict all the masked tokens, while some unmasked tokens also contribute to the loss, as detailed in Equation 4.

$$L_{mrm} = \frac{1}{m+r} \sum_{\tau \in \Gamma_m \cup \Gamma_r} CrossEntropy(\hat{\tau}, \tau) \quad (4)$$

where $\hat{\tau}$ and τ denote a predicted and ground truth token, respectively. The Γ_m and Γ_r denote the sets of masked tokens with size m and randomly selected unmasked tokens with size r , respectively.

Unlike BERT[7] which masks only 15% tokens, we vary the mask ratio during training, potentially masking all tokens to aid in predicting from scratch. We utilize linear and cosine mask schedules [3], which increases the mask ratio over time or decides randomly per iteration, respectively.

We further introduce the **Speaker-Listener Matching (SLM) Loss**, L_{slm} , to enhance the model’s understanding of speaker-listener correlation. Echoing the Image-Text Matching Loss [9], the SLM Loss employs a learnable SLM token τ^{slm} added to the input sequence, whose output helps classify the appropriateness of the listener’s reaction to the speaker, as defined in Equation 5.

$$L_{slm} = \text{CrossEntropy}(\hat{y}, y) \quad (5)$$

where $y \in \{0, 1\}$ is the label indicating whether the input listener sequence is appropriate to the speaker sequence.

2.2.2 Online Task. During online training, we utilize causal self-attention instead of bi-directional self-attention. For the online task, we train the network only by GPT-like loss [1] on all listener tokens, which we call **Causal Reaction Modeling (CRM) Loss**, as formulated in Equation 6, where d is the modality of tokens.

$$L_{crm} = \frac{1}{3T} \sum_{1 \leq t \leq T} \sum_{d \in \{AU, VA, exp\}} \text{CrossEntropy}(\hat{\tau}_t^d, \tau_t^d) \quad (6)$$

2.3 Inference

The challenge mandates generating 10 reaction sequences per speaker sequence for evaluation. We adopt LLM [1] hyper-parameters like *temperature*, *top-k*, and *top-p* for sampling control, which can empirically balance appropriateness and diversity. We also introduce *step* and *window* hyper-parameters for offline and online inference, respectively. Adopting MaskGIT’s [3] cosine sampling, the *step* determines the number of steps needed for each reaction sequence in the offline task, with *window* controlling the number of successive reaction frames generated at once in the online task. We decode the predicted tokens into the original spaces of AUs, VA, and expressions by the codebooks of tokenization.

3 EXPERIMENTS

We evaluate UniFaRN on the official dataset of the REACT2023 challenge [23][22][21][12], which is constructed from the existing UDIVA [14], RECOLA [17], and NoXI [2] datasets. Due to the page limit, a detailed description can be found in the baseline paper [22].

We follow the baseline paper [22] to evaluate our method from seven metrics: facial reaction correlation (FRCorr), appropriate facial reaction distance (FRDist), and facial reactions’ diverseness (FRDiv), facial reaction variance (FRVar), diversity among facial reactions generated from different speaker behaviours (FRDvs), synchrony between generated facial reactions and speaker behaviours (FRSyn), and facial reaction realism (FRRea). The detailed formulations can be found in the theory paper [23]. The appropriateness of reaction is measured by FRCorr and FRDist, while diversity is measured by FRDiv, FRVar, and FRDvs. Other metrics are auxiliary.

The naive baseline **Random** and **Mime** and deep learning baseline **Trans-VAE** and **BelFusion** from the baseline paper are compared.

3.1 Results

We have trained three effective models: the cosine-masked UniFaRN, finetuned UniFaRN, and 5-step UniFaRN. The **cosine-masked UniFaRN**, trained with cosine schedule of masking [3] and both online and offline losses, performs well in both tasks. The **finetuned UniFaRN**, finetuned on the cosine-masked UniFaRN using only offline losses, is better balanced for the offline task. The **5-step UniFaRN** predicted 5 frames of listener tokens at once during causal training and inference, achieving superior appropriateness in the online task on the validation set. All models used the AdamW optimizer [10] and a cosine learning rate with warmup [11]. We conduct 150k training iterations for the cosine-masked and 5-step models, and 100k for the finetuning.

The experimental results are reported in Table 1, 2, 3, and 4, where the bold, underlined, and square-bracketed numbers are the best, second-best, and third-best results, respectively. The hyper-parameters temperature, top-k, or top-p may be recorded as $a + b$, which means we use different parameters to generate the 10 reactions. For each input speaker, the i th (from 0 to 9) reaction is generated by the actual parameter $a + i * b$.

First of all, the Mime baseline, in which the listener acts as a mirror by mimicking the speaker’s facial action, appears to be very strong in terms of appropriateness. Our UniFaRN shows superiority over the deep learning baselines.

3.1.1 Offline Task. As reported in Table 1 and 2, compared with the deep learning baselines, Trans-VAE and BelFusion, the finetuned UniFaRN achieves impressive diversity on both validation and testing sets, while keeping a much higher FRCorr. These results indicate the ability of UniFaRN to model multi-modal mapping while producing diverse predictions.

Table 1 also clearly shows a trade-off between FRCorr and both FRDist and diversity which is controlled by hyper-parameters of sampling. A similar trade-off also appears in other tables. Compared to the cosine-masked UniFaRN, finetuned UniFaRN generally produces worse FRCorr but better FRDist and diversity. Comparing the sampling hyper-parameters of the cosine-masked UniFaRN, we can find out that the increase of *step* leads to higher FRCorr.

3.1.2 Online Task. As reported in Table 3 and 4, our cosine-masked UniFaRN surpass the deep learning baselines in diversity and FRCorr, while is inexplicably much worse in FRDist. The divergence of FRCorr and FRDist indicates our prediction is correlated but not similar to the ground truth, which is worth investigating. It is also perplexing that the 5-step UniFaRN achieves higher FRCorr than the cosine-masked UniFaRN on the validation set but much lower on the testing set.

4 CONCLUSION

To generate diverse while appropriate reactions in dyadic interactions, we propose the UniFaRN framework, which leverages unified layers to capture multi-modal correlations and a controllable sampling process to balance appropriateness and diversity. Experimental results show that UniFaRN achieves impressive diversity

Table 1: Results of the offline task on the validation set.

Model	Step	Temp.	Top-K	Top-P	FRCorr↑	FRDist↓	FRDiv↑	FRVar↑	FRDvs↑	FRSyc↓
Baseline Random	-	-	-	-	0.04	229.37	0.1667	0.0833	0.1667	{46.65}
Baseline Mime	-	-	-	-	0.35	91.37	0.0000	{0.0666}	0.2251	44.53
Trans-VAE	-	-	-	-	0.12	101.37	0.0265	0.0045	0.0322	49.000
BeLFusion ($k=10$)	-	-	-	-	0.14	94.26	0.0134	0.0078	0.0149	[46.94]
cosine-masked UniFaRN	10	1	5	0.33	{0.265893}	102.586991	0.005398	0.046844	0.164840	49.00
cosine-masked UniFaRN	8	0.912-0.08	5	0.05+0.1	[0.259561]	102.800089	0.021097	0.046578	0.165070	49.00
cosine-masked UniFaRN	1	0.33	5	1	0.221537	95.166077	0.051129	0.052342	0.166786	49.00
finetuned UniFaRN	1	0.33	5	1	0.213486	93.554599	0.05798	0.057007	[0.199116]	48.781883
finetuned UniFaRN	1	0.33	6	1	0.216819	{92.699827}	[0.059412]	0.056588	0.196448	48.74255
finetuned UniFaRN	1	0.5	8	1	0.200294	[92.728297]	{0.082844}	[0.061665]	{0.200131}	48.528606

Table 2: Results of offline task on the testing set.

Model	Step	Temp.	Top-K	Top-P	FRCorr↑	FRDist↓	FRDiv↑	FRVar↑	FRDvs↑	FRRea↓	FRSyc↓
Baseline Random	-	-	-	-	0.04	237.62	0.1667	0.0833	0.1667	-	{43.99}
Baseline Mime	-	-	-	-	0.38	{92.95}	0.0000	[0.0723]	0.2474	-	38.66
Trans-VAE	-	-	-	-	0.10	98.48	0.0242	0.0040	0.0263	69.24	[44.88]
BeLFusion ($k=10$)	-	-	-	-	0.13	89.84	0.0137	0.0078	0.0149	-	45.02
cosine-masked UniFaRN	8	0.912-0.08	5	0.05+0.1	{0.21803}	119.32016	0.02812	0.05854	0.15287	-	48.99065
finetuned UniFaRN	1	0.33	5	1	[0.15062]	98.77017	0.07457	0.06795	0.17756	166.6469	46.29724
finetuned UniFaRN	1	0.33	6	1	0.14937	98.10896	[0.07729]	0.06839	[0.17811]	166.676	46.40054
finetuned UniFaRN	1	0.5	8	1	0.13967	[97.84649]	{0.10445}	{0.07361}	{0.18493}	-	46.13891

Table 3: Results of online task on the validation set.

Model	Window	Temp.	Top-K	Top-P	FRCorr↑	FRDist↓	FRDiv↑	FRVar↑	FRDvs↑	FRSyc↓
Baseline Random	-	-	-	-	0.04	229.37	0.1667	[0.0833]	0.1667	{46.65}
Baseline Mime	-	-	-	-	0.35	91.37	0.0000	0.0666	{0.2251}	44.53
Trans-VAE	-	-	-	-	0.14	134.40	[0.1149]	0.0594	0.1224	[46.81]
BeLFusion ($k=10$)	-	-	-	-	0.14	{93.64}	0.0131	0.0076	0.0143	47.10
cosine-masked UniFaRN	1	1	5	0.33	0.173723	125.744884	{0.134123}	0.091884	0.230472	47.940929
cosine-masked UniFaRN	1	1	0	0	0.179775	126.515034	0	{0.087294}	[0.224944]	48.4224
cosine-masked UniFaRN	2	0.912	3	0.317	0.17618	125.179784	0.015088	0.080431	0.215069	48.175419
5-step UniFaRN	5	1	5	0.33	[0.198886]	[102.552788]	0.053792	0.06612	0.103502	49.00
5-step UniFaRN	5	0.33	5	1	{0.208453}	103.636494	0.074614	0.065093	0.105629	49.00

Table 4: Results of online task on the testing set.

Model	Window	Temp.	Top-K	Top-P	FRCorr↑	FRDist↓	FRDiv↑	FRVar↑	FRDvs↑	FRRea↓	FRSyc↓
Baseline Random	-	-	-	-	0.04	237.62	0.1667	0.0833	0.1667	-	{43.99}
Baseline Mime	-	-	-	-	0.38	{92.95}	0.0000	0.0723	0.2474	-	38.66
Trans-VAE	-	-	-	-	0.13	135.57	[0.1168]	0.0604	0.1202	71.15	[44.31]
BeLFusion ($k=10$)	-	-	-	-	0.13	89.42	0.0133	0.0077	0.0143	-	44.80
cosine-masked UniFaRN	1	1+0.022	1+1	0.2	{0.18346}	124.59443	0.08981	[0.07614]	[0.22324]	-	48.24406
cosine-masked UniFaRN	1	1	5	0.33	[0.17580]	124.55294	{0.13604}	{0.08083}	{0.22617}	-	47.82080
5-step UniFaRN	5	1	0	0.33	0.12388	100.33885	0.07015	0.06321	0.11253	-	48.99101
5-step UniFaRN	5	0.33	5	1	0.13468	[99.98538]	0.07650	0.06334	0.11146	167.8195	49.00

while keeping good appropriateness. As a result, we won first place in the REACT2023 challenge.

Although we have won the challenge, quite a lot of opportunities for further investigation emerge, including feature selection, model

scalability measurement, and sampling method selection. Among them, the effect of the *step* parameter trouble us particularly. After all, the Mime baseline is still calling for challengers.

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